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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network sudo**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM

2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SHH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 83386sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

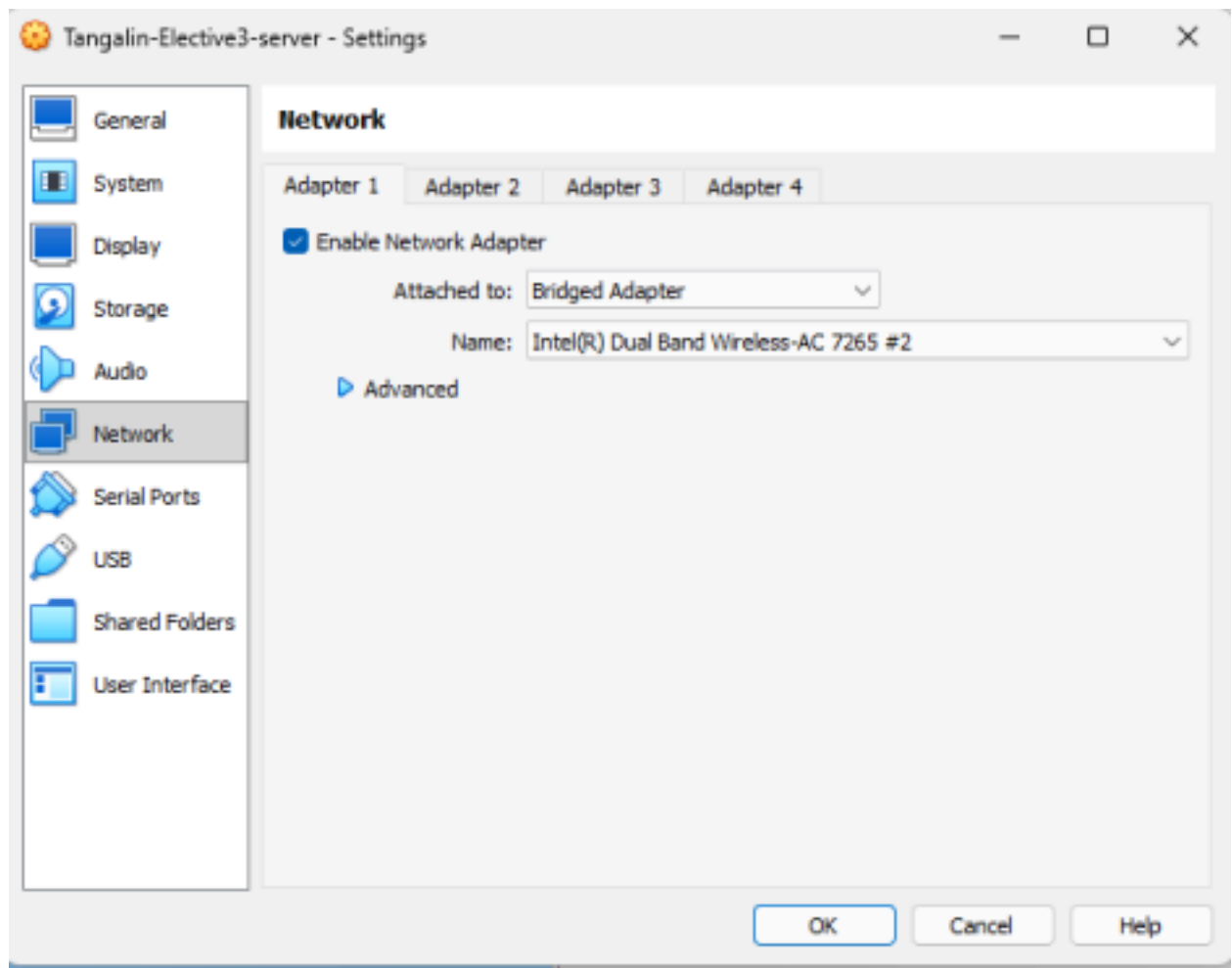
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

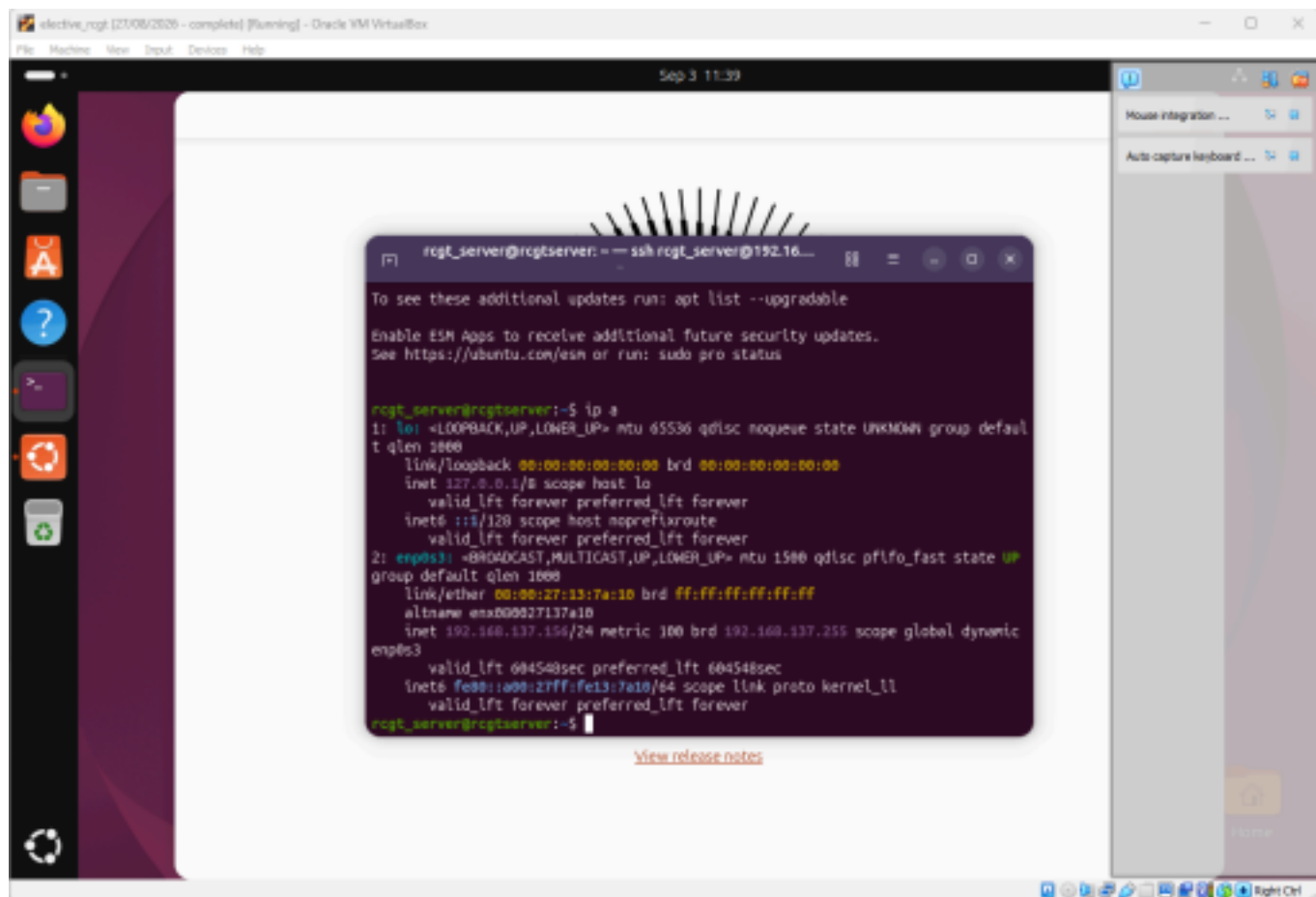
Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
 - I changed the attached adaptor from NAT to Bridged Adapter in Adaptor 1.



- Server IP address (output of ip a or hostname -l)
 - I run the command “ip a” to get the IP address of the server and at the desktop.



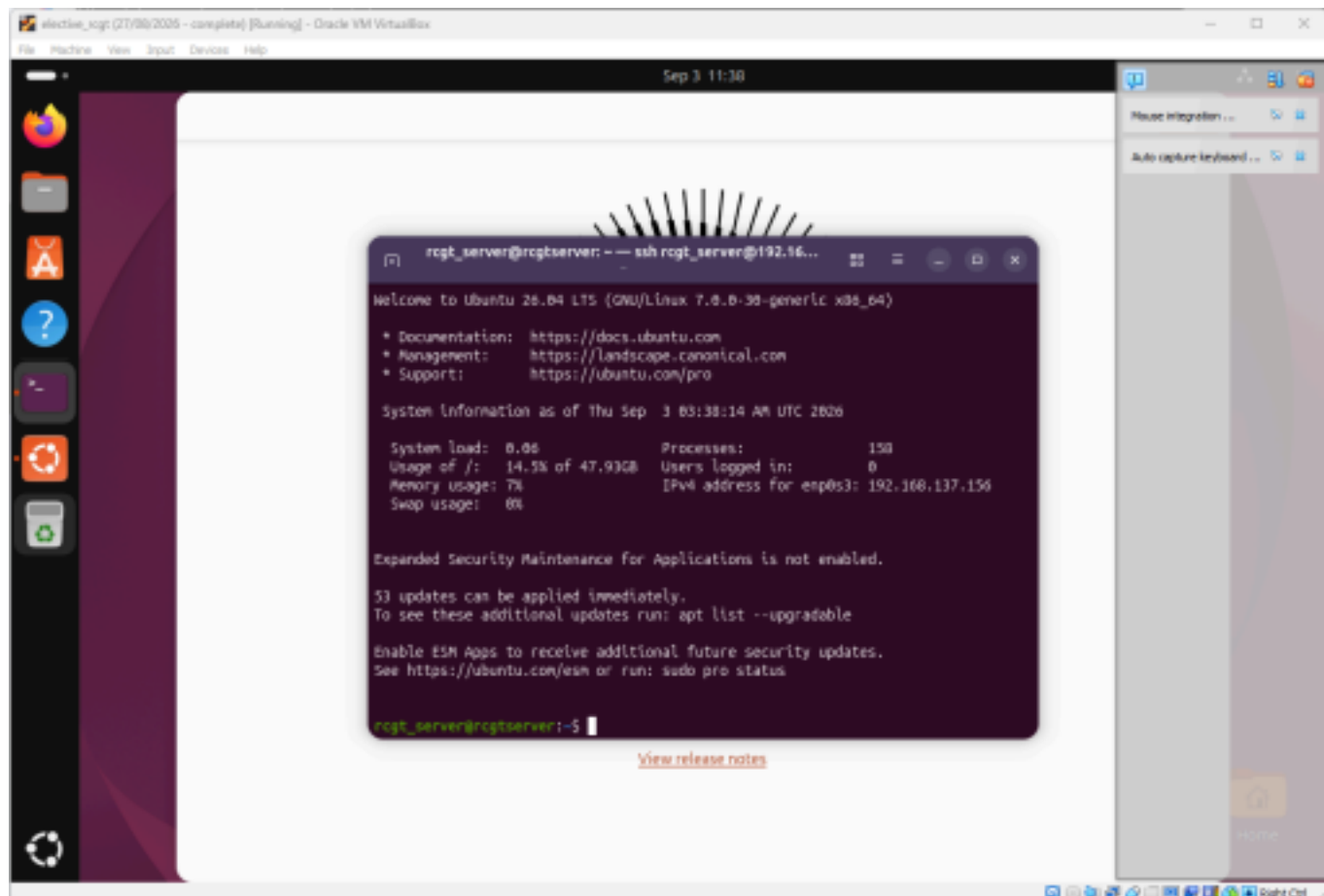
```
rcgt_server@rcgtserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:00:27:13:7a:10 brd ff:ff:ff:ff:ff:ff
    altname enx000027137a10
    inet 192.168.137.156/24 metric 100 brd 192.168.137.255 scope global dynamic enp0s3
        valid_lft 604784sec preferred_lft 604784sec
    inet6 fe80::a00:27ff:fe13:7a10/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
rcgt_server@rcgtserver:~$
```

- SSH service status (sudo systemctl status ssh)
 - I run the command to see the status.

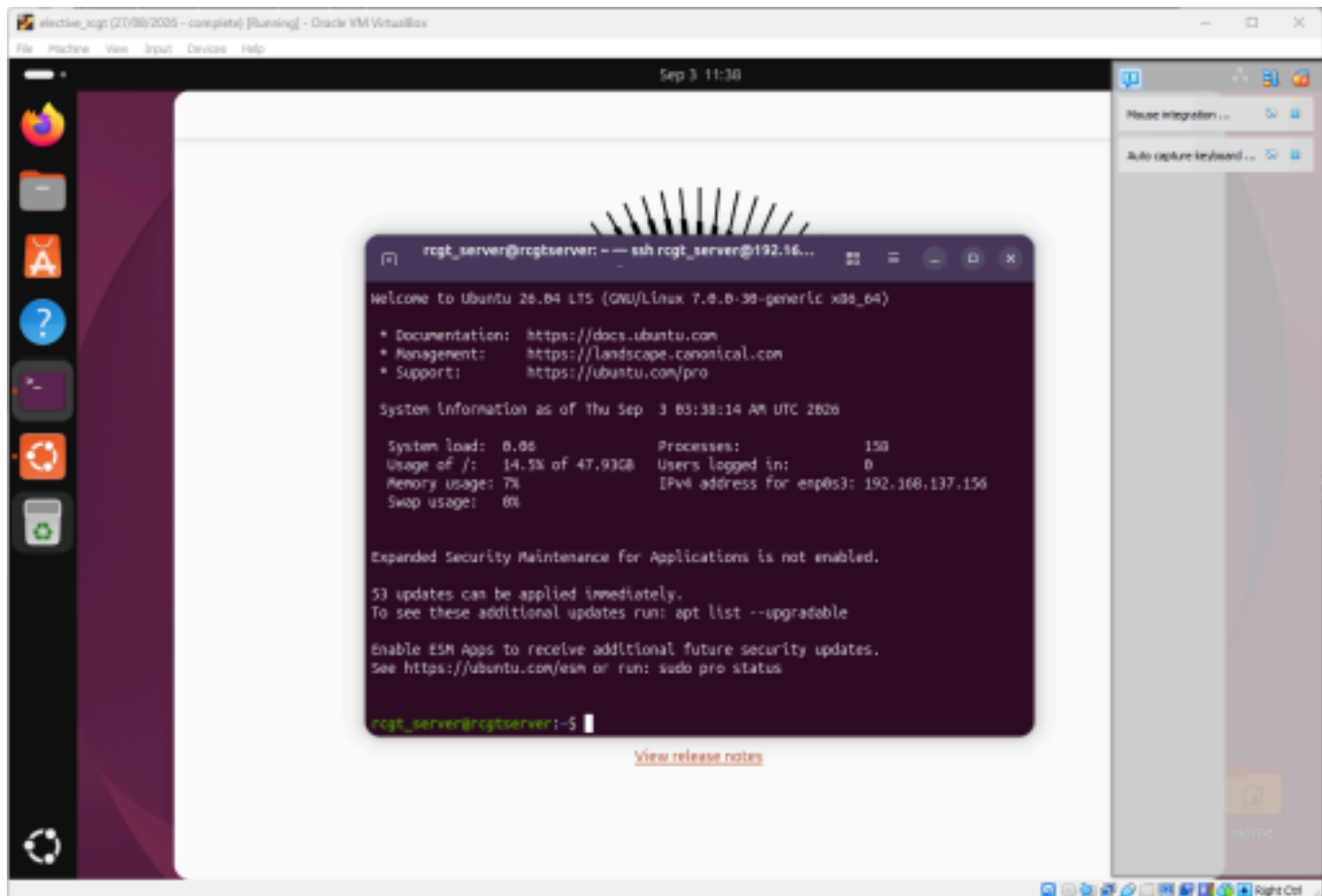
```
rcgt_server@rcgtserver:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-09-03 03:34:59 UTC; 13min ago
   Invocation: e5b01b03c95b4be099b34a91119ff356
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 1157 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1194 (sshd)
     Tasks: 1 (limit: 1567)
    Memory: 9M (peak: 10.9M)
       CPU: 97ms
   CGroup: /system.slice/ssh.service
           └─1194 "sshd: /usr/sbin/sshd -O [listener] 0 of 10-100 startups"

Sep 03 03:34:59 rcgtserver systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 03 03:34:59 rcgtserver sshd[1194]: Server listening on 0.0.0.0 port 22.
Sep 03 03:34:59 rcgtserver sshd[1194]: Server listening on :: port 22.
Sep 03 03:34:59 rcgtserver systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Sep 03 03:38:14 rcgtserver sshd-session[1474]: Accepted password for rcgt_server from 192.168.137.170 port 61164 ssh2
Sep 03 03:38:14 rcgtserver sshd-session[1474]: pam_unix(sshd:session): session opened for user rcgt_server(uid=1000) by rcgt_server(uid=0)
Sep 03 03:47:06 rcgtserver sshd-session[1586]: Accepted password for rcgt_server from 192.168.137.156 port 34852 ssh2
Sep 03 03:47:06 rcgtserver sshd-session[1586]: pam_unix(sshd:session): session opened for user rcgt_server(uid=1000) by rcgt_server(uid=0)
rcgt_server@rcgtserver:~$
```

- Successful SSH connection from host
 - I enter the server ip address and managed to connect it from the host.



- Verification commands executed inside VM
 - I run the command "ssh rcgt_server@192.168.137.156" to verify.



2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSCS 3rd Year

Date Submitted: September 03, 2026

Instructor: Johneros P. Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☒ Create and configure a Virtual Machine with Ubuntu Server
- ☒ Configure Bridged Network Adapter for network access
- ☒ Determine and verify the server's IP address
- ☒ Install OpenSSH Server and enable the service
- ☒ Establish successful SSH connection from host machine
- ☒ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu
Virtualization Software (VirtualBox/VMware)	VirtualBox
Ubuntu Server Version	26.04
VM Name	Tangalin-Elective3-server
RAM Allocated	4000 MB
Storage Allocated	100 GB
Network Mode	Bridged Adapter
SSH Client Used	OpenSSH

3 VM CONFIGURATION

Setting	Value
VM Name	Tangalin-Elective3-server
OS Type	Ubuntu (64-bit)
Memory	4000 MB
Hard Disk	100GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

Server IP Address: 192.168.137.156

Network Interface: Physical

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.168.137.156 rcgt_server
Username	

